



Fig. 3: Various connections and modifications in PSU outputs

are meant for connections to the motherboard, the connections need to be slightly modified for use of the unit as a bench power supply.

All the required connections for fixed but multiple outputs are shown in Fig. 3 and the author's prototype of the same is shown in Fig. 4. Front panel of the author's prototype is shown in Fig. 5.

Power-on switch. The green wire in ATX is PS_ON or 'power supply on' wire. It is used by the motherboard to turn on the PSU. Normally, its voltage level is +5V DC. To turn on the PSU, it needs to be grounded. So you need to connect an on/off switch between PS_ON and ground.

Standby indicator. When AC power cord is plugged into 230V AC socket and power switch is in 'off' position, the PSU is in standby mode. To provide an indication for standby mode, connect an LED to purple wire, which provides a 5VSB signal from ATX PSU. 5VSB is active-high when the PSU is in standby mode. The voltage level of 5VSB is +5V DC, so you need to connect a 220-ohm, 0.25-watt resistor in series with the LED. As soon as power switch is operated to power-on the PSU, 5VSB goes low and the LED stops glowing, indicating that the PSU is

in power-on mode.

Power-on indicator. For power-on indication, you need to connect an LED to grey wire through a 220-ohm, 0.25-watt resistor. This grey wire provides PGS or power-good signal. In standby mode, this signal is active-low. After powering up the PSU, it becomes active-high (+5V DC).

Series fuse. You may ask what's the need of a series fuse when the PSU has a very good inbuilt short-circuit protection. The fuse is used here not to protect the PSU but your valuable project.

As you know, the current rating of the PSU is moderately high at more than 10 amperes. But in general, an average electronics project or circuit doesn't need such high current. If there is any short circuit in your project, the short-circuit current will reduce to the level of the fuse current. Without a fuse, the short-circuit current will reach the maximum output current of the PSU, severely damaging the circuit.

To make your power supply more effective, remove the PCB board from the PSU enclosure. Place it in a larger enclosure and mount banana sockets for different outputs. You can also use an ammeter in series with the output to



Fig. 4: Author's prototype with internal connections



Fig. 5: Front panel of the author's prototype

PARTS LIST-1 FOR FIG. 3

Semiconductors:

LED1, LED2 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2 - 220-ohm

Miscellaneous:

- ATX power supply unit (PSU)
- 1A fuse with holder
- Five banana sockets
- AC power cord for PSU

PARTS LIST-2 FOR FIG. 6

Semiconductors:

IC1 - LM317 adjustable voltage regulator

D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 120-ohm

VR1 - 4.7-kilo-ohm potentiometer

Capacitors:

C1 - 100 μ F, 40V electrolytic

C2 - 10 μ F, 40V electrolytic

Miscellaneous:

- +12V from ATX
- -12V from ATX
- Heat-sink for LM317
- Banana sockets